

Hima Shanthi Rayapati

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Education

Arizona State University, Data Science, Analytics and Engineering (Human-Centered Applications)	08/2026 – 12/2027
Arizona State University, BS in Psychology and Mathematics (Statistics)	01/2023 – 05/2026
• Minor: Philosophy GPA: 4.0/4.0	

Skills

- **Programming & Statistical Tools:** R, SAS, MATLAB, Python, JMP, JASP, SQL (Microsoft Access), Java, C, Bash, REST APIs
- **Machine Learning & NLP (Foundational):** Principal Component Analysis (PCA), Neural network basics, Supervised vs. Unsupervised learning, Model evaluation concepts, Introductory text analysis (Voyant Tools)
- **Data Analysis & Visualization:** Tableau, Excel, Google Sheets, Alteryx
- **Experimental & Research Tools:** jsPsych, PsychoPy, JavaScript, HTML, CSS, Qualtrics, SurveyMonkey
- **Collaboration & Productivity:** Slack, Teams, Zoom, Word, Outlook, Adobe PowerPoint, OneDrive, Dropbox

Projects

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| Trust in Advice: AI vs. Human Sources (Data Analysis & Written Report) | <i>Trust in AI.com/2025</i> |
| <ul style="list-style-type: none">• Analyzed 30,000+ human-AI interactions to study effects of advice source.• Modeled decision changes before and after advice exposure.• Found advice correctness as primary driver, with modest AI-related effects. | |

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| Analyzing the Impact of AI on Stack Overflow Participation (NLP & Data Analysis) | <i>GitHub</i> |
| <ul style="list-style-type: none">• Analyzed large-scale Stack Overflow data using Python to study changes in user behavior after AI tool releases.• Built time-series datasets and computed engagement metrics (answer rates, response time, retention) using Python.• Performed statistical comparisons and visualized platform-level trends.• Explored tag-based trends and basic NLP features to examine topic shifts. | |

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| Predictive Analysis of Team Performance in Baseball using SAS | <i>Baseball Analysis.com/2024</i> |
| <ul style="list-style-type: none">• Developed a statistical model to predict team wins using historical baseball data.• Conducted EDA and correlation analysis on Minnesota Twins batting records (1998-2024, excluding 2020).• Built multiple regression models to identify key predictors of team wins, selecting runs and batting averages for a parsimonious model.• Addressed multicollinearity and validated model assumptions to improve predictive accuracy. | |

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| Predicting Daily Stock Prices of ADP using SAS | <i>ADP Stock.com/2024</i> |
| <ul style="list-style-type: none">• Created a predictive model for ADP stock prices based on financial indicators.• Applied EDA, correlation analysis, and multiple regression (stepwise, backward, forward selection) to determine significant predictors.• Developed a quadratic regression model with strong explanatory power while addressing multicollinearity.• Examined sector-specific dynamics and market trends influencing ADP's stock behavior. | |

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| Stochastic Dual-Timescale Model of Learning (ODE & SDE Modeling) | <i>Final Project / 2025</i> |
| <ul style="list-style-type: none">• Developed differential equation model of fast and slow learning processes.• Extended to stochastic framework to examine effects of noise.• Verified theoretical properties via simulation.• Linked model behavior to memory retention and learning variability. | |

Probabilistic Reward Learning Task using PsychoPy

Reward Learning.com/2025

- Designed probabilistic reward-learning task with informative vs. non-informative features.
- Examined learning under uncertainty and perceptual noise.
- Collected and analyzed participant data using Python.

Additional projects and full write-ups available at my portfolio.

Research Experience

Research Assistant, *Thinking Across Languages and Contexts Lab*, ASU

08/2025 – Present

- Assist a graduate student with testing experiments built in jsPsych and JavaScript.
- Help check whether data are recorded correctly and organized for analysis.
- Perform exploratory data analysis and assist in interpreting experimental results under supervision.

Research Assistant, *Perception, Action, Language & Meaning Lab*, ASU

08/2025 – Present

- Run two-person collaborative task sessions (e.g., Taboo), observing and recording coordination, communication, and performance data using VR, motion-capture, and eye-tracking.
- Work with the lab's time-series behavioral data, learning how to clean, organize, and explore it.
- Read papers and explore possible methods (e.g., multilevel vector autoregression) to understand which approaches might fit the lab's data.
- Contribute to discussions by sharing preliminary notes and observations from trial analyses.

Volunteer Research Assistant, *Longitudinal & Behavioral Data Science Lab*, ASU

08/2024 – 12/2024

- Participated in bi-weekly lab meetings, contributing to discussions and staying updated on research progress.

Research Assistant, *Evolution, Ecology, and Social Behavior Lab*, ASU

08/2023 – 11/2023

- Assisted with survey design, pilot testing, data coding, and literature reviews in social cognition research.
- Participated in lab meetings and contributed to discussions on study design and interpretation.

Work Experience

Research Aide, ASU – Tempe, AZ

12/2024 – Present

- Review, categorize, and structure user-generated data to ensure quality and consistency.
- Assign relevant tags to comments for improved organization and searchability.
- Evaluate AI-generated tags against human annotation standards, analyzing human-AI agreement and error patterns across prompt versions to guide model refinement.
- Perform data cleaning tasks, identifying and correcting inconsistencies to maintain data accuracy.
- Work with databases to structure data and ensure accessibility for future use.

Lead Survey Assistant, ASU – Tempe, AZ

05/2023 – Present

- Conduct phone-based survey interviews with currently employed and recently graduated ASU students, ensuring accurate and comprehensive data collection by clarifying responses and addressing any ambiguities.
- Assist in the hiring process, including interviewing and evaluating candidates.
- Perform data cleaning and ensure the integrity of datasets for analysis.
- Train new student workers in survey procedures and oversee testing and calibration of survey tools.

Intern, Public Safety Crisis Solution – Phoenix, AZ

08/2024 – 11/2024

- Performed quantitative analysis to support research, program development, and accreditation documentation.
- Contributed research-based content on clinical procedures addressing first responders' mental health initiatives.

Advising Office Aide, ASU – Tempe, AZ

08/2023 – 05/2024

- Supported advising operations through scheduling, communication, and administrative workflows.
- Managed data entry, marketing/outreach materials, and front-desk support for students and staff.